



GTEM

Global Tsunami Evacuation Model

User Manual

Version 1.0.0 · August 2026

An agent-based model of pedestrian tsunami evacuation, written for staff in coastal local governments rather than for modelling specialists.



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Funded by the Coalition for Disaster Resilient Infrastructure
CDRI Fellowship Programme 2025–2026

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Before you begin

0.1 Who this manual is for

You do not need to be a programmer or a modeller. The manual assumes you can install software, open a terminal window, and edit a text file. Everything else is explained.

If you have used a geographic information system such as QGIS, Section 4 will make more sense — but you can run GTEM on the areas supplied with it without touching a GIS at all.

0.2 What GTEM does

You give GTEM three things:

1. a road network for a coastal area,
2. where people are, and
3. the number of minutes between the earthquake and the arrival of the wave.

GTEM simulates every person walking to the nearest safe zone and reports who reaches safety in time, who does not, **where they were when they ran out of time**, and which streets became bottlenecks.

0.3 What GTEM is for, and what it is not for

Read this before you use a result in a decision

GTEM is a tool for **comparing options**, not for predicting what will happen.

“Opening this road leaves 400 fewer people exposed” is a defensible use of the model. “1,847 people will die” is not.

GTEM has **not been validated against an observed evacuation**. Until it has, treat every absolute number as provisional. The full statement of what the model can and cannot answer is in `docs/LIMITATIONS.md`, and you should read it before presenting results to anyone.

0.4 How to use this manual

If you want to...	Read
install GTEM and see it work	Section 1
understand what the results mean	Section 2
run realistic studies	Section 3
use GTEM on your own town	Section 4
know how the model works inside	Section 5
look something up	Section 6

A note on commands

Anything in a dark box is typed into a terminal. Type the command, press Enter, and compare what you see with what the manual shows. Text after a `#` in a configuration file is a comment and is ignored.

Part 1

Getting started

1.1 What you need

Item	Notes
A computer running Windows 10/11, macOS or Linux	Any machine from the last few years is enough. A full run of a 17,000-person area takes about 90 seconds.
NetLogo 7.0.4	Free. This is the simulation engine GTEM drives. Java is bundled with it — you do not need to install Java separately.
Miniforge, Miniconda or Anaconda	Free. Used to install the Python packages GTEM needs, without disturbing anything else on your machine.
About 1 GB of free disk space	The download is small; results accumulate.

1.2 Installing

1.2.1 Step 1 — Install NetLogo

Download NetLogo 7.0.4 from ccl.northwestern.edu/netlogo and install it in the default location.

System	Default location GTEM looks in
Windows	C:\Program Files\NetLogo 7.0.4
macOS	/Applications/NetLogo 7.0.4
Linux	/opt/netlogo-7.0.4

If you install it somewhere else, set an environment variable named `NETLOGO_HOME` to that folder. GTEM will tell you if it cannot find NetLogo, and will list the places it looked.

SCREENSHOT S1 — to be added

Capture: The NetLogo download page, with the 7.0.4 download button visible.

Why: New users often download the newest version rather than 7.0.4. A picture of the right button prevents the most common installation mistake.

Save as `figures/S1.png` and replace this box with `#figure(image("figures/S1.png"), caption: [...])` — see the appendix at the end of this manual.

1.2.2 Step 2 — Get GTEM

Download or clone the GTEM folder and put it somewhere with a short path — for example `C:\GTEM` on Windows, or your home folder on macOS or Linux.

Avoid long folder paths on Windows

Windows has a 260-character limit on file paths. GTEM writes results into nested folders, so placing it inside a deeply nested location such as `C:\Users\...\OneDrive\Documents\Projects\2026\...` can cause errors that look unrelated. A short path avoids the problem entirely.

SCREENSHOT S2 — to be added

Capture: The unzipped GTEM folder open in Windows Explorer or macOS Finder, showing `main.py`, `data`, `examples`, `docs` and `src`.

Why: Confirms the reader has unzipped to the right level. A common error is a folder containing only another folder.

Save as `figures/S2.png` and replace this box with `#figure(image("figures/S2.png"), caption: [...])` — see the appendix at the end of this manual.

1.2.3 Step 3 — Create the environment

Open a terminal — on Windows this is the **Miniforge Prompt** or **Anaconda Prompt** from the Start menu, not the ordinary Command Prompt. Move into the GTEM folder and run:

```
conda env create -f environment.yml
conda activate gtem
```

The first command downloads and installs everything GTEM needs and takes a few minutes. The second switches your terminal into that environment.

You must activate the environment in every new terminal

`conda activate gtem` applies only to the window you type it in. If you close the terminal and open a new one, run it again. If a command suddenly reports that a package is missing, this is almost always the reason.

SCREENSHOT S3 — to be added

Capture: The Miniforge Prompt on Windows immediately after `conda activate gtem`, with the prompt showing `(gtem)` at the start of the line.

Why: The `(gtem)` prefix is the visual confirmation that the environment is active. Readers who miss it hit confusing errors later.

Save as `figures/S3.png` and replace this box with `#figure(image("figures/S3.png"), caption: [...])` — see the appendix at the end of this manual.

1.3 Checking that it worked

Run:

```
python check_environment.py
```

This imports every package GTEM needs, finds NetLogo, and confirms the model and data are present. You should see something close to this:

```
GTEM 1.0.0 - environment check
platform: Darwin 25.6.0 (arm64)
python: 3.10.6
tested against: NetLogo 7.0.4, Java 17.0.2 (bundled with NetLogo 7.0.4)

[ OK ] numpy      2.2.6      (numerics)
[ OK ] pandas     2.3.3      (tables and CSV output)
[ OK ] matplotlib 3.10.9     (figures)
[ OK ] networkx   2.8.7      (route pre-computation)
[ OK ] pynetlogo  0.5.2      (the bridge to the NetLogo engine)
[ OK ] jpype      1.5.0      (starting the Java virtual machine)
[ OK ] fpdf       1.7.2      (the PDF report)
[ OK ] geopandas  1.1.1      (input checking (check_inputs.py))
[ OK ] shapely    2.1.2      (geometry)
[ OK ] pyproj     3.7.1      (coordinate reference systems)
[ OK ] pytest     9.1.1      (the test suite (developers only))
[ OK ] NetLogo    /Applications/NetLogo 7.0.4
[ OK ] JVM        libjvm.dylib
[ OK ] model      gtem_model.nlogox
[ OK ] data folder data

Everything needed is present. GTEM can run on this machine.
Next step: python main.py --config examples/config_example.txt
```

Terminal 1: `check_environment.py` on a correctly installed machine.

If anything is wrong, the tool prints the exact command to fix it and exits with an error. It checks by **importing** each package rather than merely looking for it on disk, because a package can be present and still fail to load.

1.4 Your first run

Start with the built-in test area `Synthetic_Corridor`. It is deliberately artificial: two straight 1,000-metre corridors, each ending at a safe zone. A person walking at the free-flow speed of 1.33 m/s covers 1,000 m in **12.53 minutes** — so you know in advance what the answer should be.

Create a file called `my_first_run.txt` containing:

```
zone           = Synthetic_Corridor
adults         = 50
elderly        = 0
children       = 0
tsunami_eta    = 30
departure_mean = 0
dt             = 5
seed           = 1
recompute_routes = true
```

Then run:

```
python main.py --config my_first_run.txt
```

The run finishes in under a minute and ends with everyone safe at about 12.5 minutes — matching the hand calculation. If you see that, GTEM is working correctly on your machine.

Results are written to `Outputs/Synthetic_Corridor/1/`. Open the PDF first.

SCREENSHOT S4 — to be added

Capture: The `Outputs/Synthetic_Corridor/1/` folder showing the 14 output files.

Why: Shows the reader where results appear and what a complete set looks like, so they can tell at a glance if something is missing.

Save as `figures/S4.png` and replace this box with `#figure(image("figures/S4.png"), caption: [...])` — see the appendix at the end of this manual.

1.5 A real area

Now run one of the supplied Peruvian study areas. First check its data:

`python check_inputs.py Chimbote_Zonal`

```
Checking /home/you/gtem/data/Chimbote_Zonal

[ OK ] zone boundary      Chimbote_Zonal.shp
[ OK ] road network      rutas_Chimbote_Zonal.shp
[ OK ] intersections     puntos_Chimbote_Zonal.shp
[ OK ] census blocks     manzanas_Chimbote_Zonal.shp

[ OK ] Chimbote_Zonal.shp      CRS WGS 84 / UTM zone 17S
[ OK ] rutas_Chimbote_Zonal.shp CRS WGS 84 / UTM zone 17S
[ OK ] puntos_Chimbote_Zonal.shp CRS WGS 84 / UTM zone 17S
[ OK ] manzanas_Chimbote_Zonal.shp CRS WGS 84 / UTM zone 17S

[ OK ] rutas_Chimbote_Zonal.shp: has 'start_node'
[ OK ] rutas_Chimbote_Zonal.shp: has 'end_node'
[ OK ] rutas_Chimbote_Zonal.shp: has 'cost'
[ OK ] rutas_Chimbote_Zonal.shp: has 'lanes'
[ OK ] puntos_Chimbote_Zonal.shp: has 'fid'
[ OK ] puntos_Chimbote_Zonal.shp: has 'is_shelter'
[ OK ] census blocks: population field 'T_TOTAL', total 42,075 over 388 blocks
[WARN] 8 block(s) have a missing or non-numeric 'T_TOTAL'; they will receive no population

[ OK ] network: 3621 nodes, 4046 links, 5 safe zone(s)
[WARN] network is in 2 disconnected pieces
[WARN] 884 of 4468 starting points (19.8%) cannot reach ANY safe zone
[WARN] 32 link(s) shorter than 1 m (min 0.015 m) will produce implausible densities

Usable, with 4 warning(s). Read them before trusting results.
```

Terminal 2: Checking a real study area. Warnings are normal — real data is rarely perfect. What matters is that you read them.

Then run it:

`python main.py --config examples/config_example.txt`

This simulates 17,261 people with a 23-minute tsunami arrival time, and takes about 90 seconds.

Part 2

Reading your results

Every run writes fourteen files into `Outputs/<area>/<seed>/`. Start with the PDF; the CSV files are there when you need the numbers behind it.

2.1 The four numbers that matter

They appear on page 1 of the PDF and in `Run_Summary.csv`.

Number	What it means
Evacuated before the tsunami	Reached a safe zone before the wave. These are the only people who are actually safe.
NOT evacuated	Everybody else. Always look at this number, not only the first one.
Caught in transit	Still walking when the wave arrived. Usually improved by earlier departure or by placing safe zones closer.
Stranded (no route)	No path to any safe zone from where they started — before the simulation even began. This is a network problem, not a behaviour problem. No amount of preparedness or warning time fixes it.

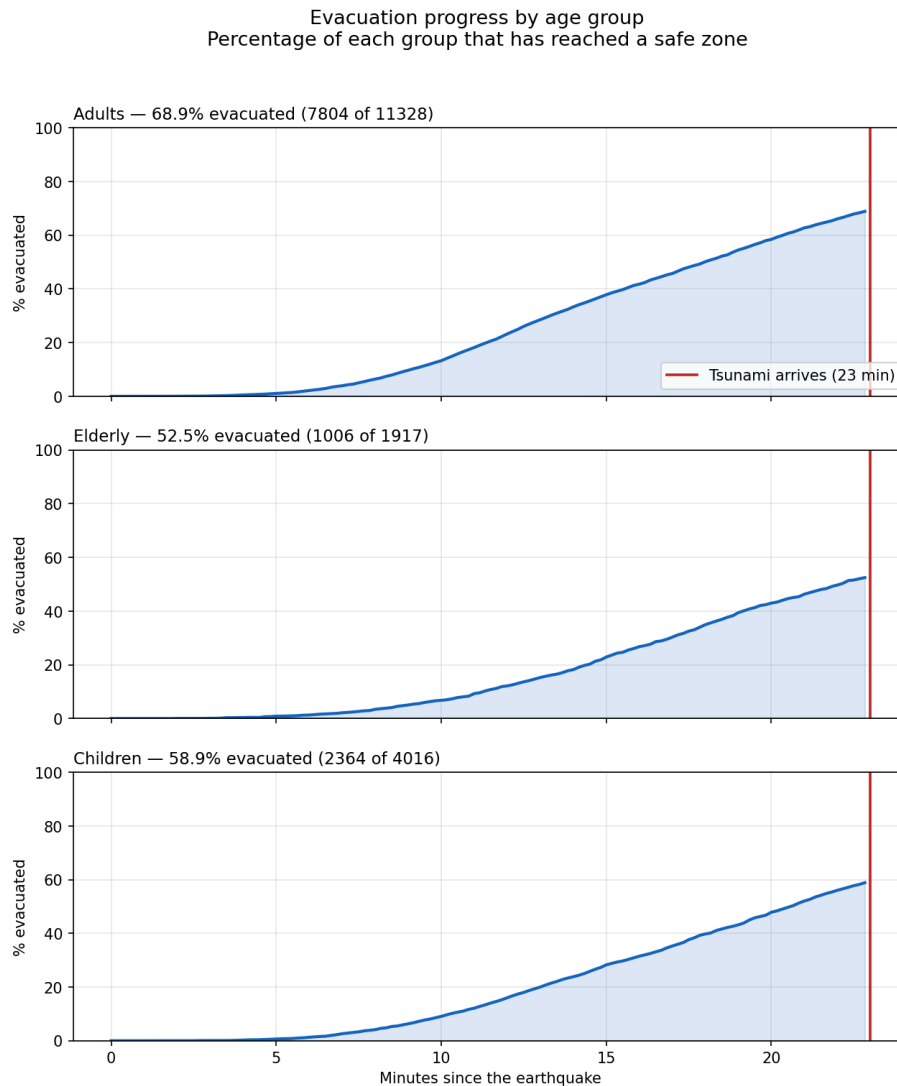
The three outcomes always add up to the number of people simulated. GTEM enforces this while it runs: if they ever failed to add up, the run would stop with an error rather than report a number that leaves people out.

Reaching a safe zone after the wave is not safety

GTEM stops the simulation at the tsunami arrival time. Anyone still walking is counted as **not evacuated**, however close they were. This is deliberate. A model that counts late arrivals as survivors flatters the plan being tested.

2.2 The five figures

2.2.1 Figure 1 — Evacuation progress by age group



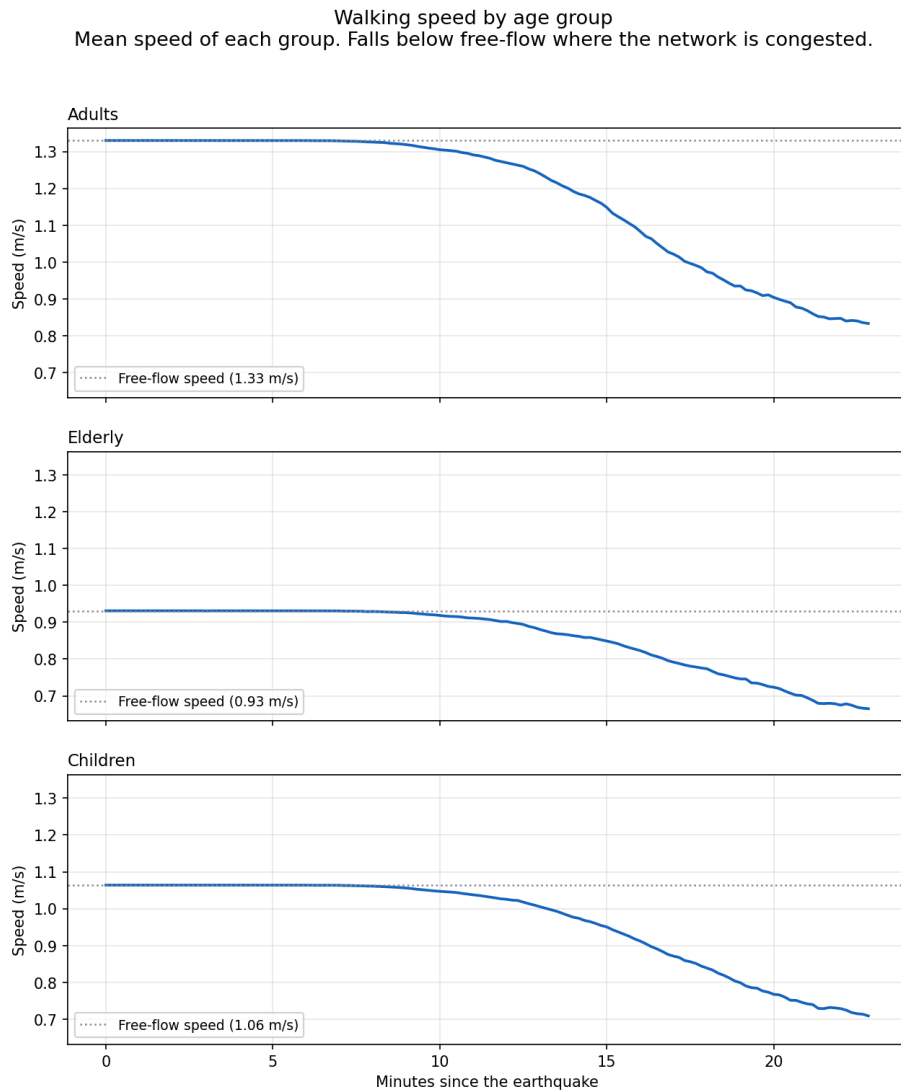
GTEM 1.0.0

Figure 1: Each age group has its own panel, shown as a percentage of that group. The vertical red line is the tsunami arrival time.

What to look for. Compare the three panels. In the example above, adults reach 68.9% but the elderly reach only 52.5% — a 16-point gap. That gap is an actionable finding: it points to assisted evacuation or targeted outreach, not to more signage.

Anything to the right of the red line did not happen in time.

2.2.2 Figure 2 — Walking speed by age group



GTEM 1.0.0

Figure 2: Mean walking speed for each group. The dotted line is that group's free-flow speed — the speed with the street to themselves.

What to look for. A curve sitting on the dotted line means people are walking freely. A curve falling away from it means crowding. In the example, all three groups drop sharply after about minute 12, which is congestion building on the approaches to the safe zones, not people becoming tired — GTEM has no fatigue.

2.2.3 Figure 3 — Vulnerability by starting location



GTEM 1.0.0

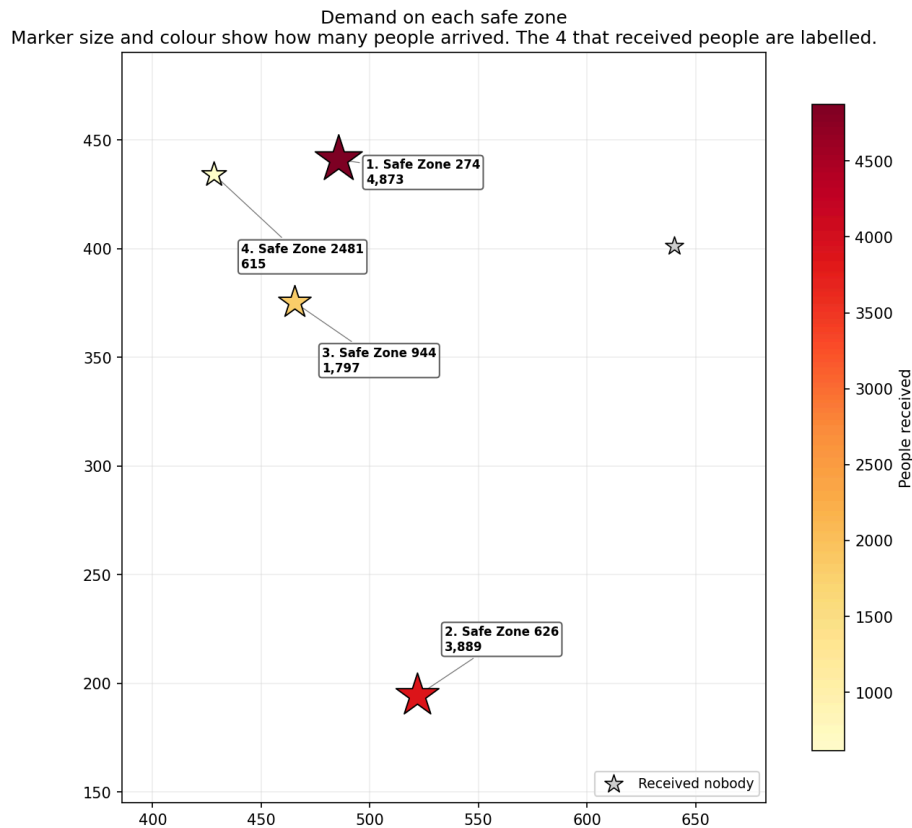
Figure 3: Where each person **started**, coloured by how long they took to reach safety. Black points never reached safety at all.

What to look for. Black clusters. These are the priority areas: places where evacuation **fails**, not merely where it is slow. In the example the whole south-western district is black.

This map shows everybody, including the people who did not make it

A blank area on this map means nobody started there. It does **not** mean everybody there was safe. Places where evacuation failed are drawn in black precisely so that they cannot be mistaken for empty space.

2.2.4 Figure 4 — Demand on each safe zone



GTEM 1.0.0

Figure 4: Marker size and colour show how many people arrived. The busiest are labelled.

What to look for. Two things.

First, **concentration**. In the example one safe zone receives 4,873 people while another receives none. GTEM sends everyone to the nearest safe zone by distance, so demand piles up wherever that rule points.

Second, **capacity**. GTEM assumes safe zones are infinitely large. It will not warn you that a plaza rated for 500 people received 1,200. Checking the numbers on this figure against the real capacity of each site is a manual step, and an important one.

A safe zone that receives nobody is worth investigating: it may be unreachable, badly placed, or simply redundant.

2.2.5 Figure 5 — Street congestion

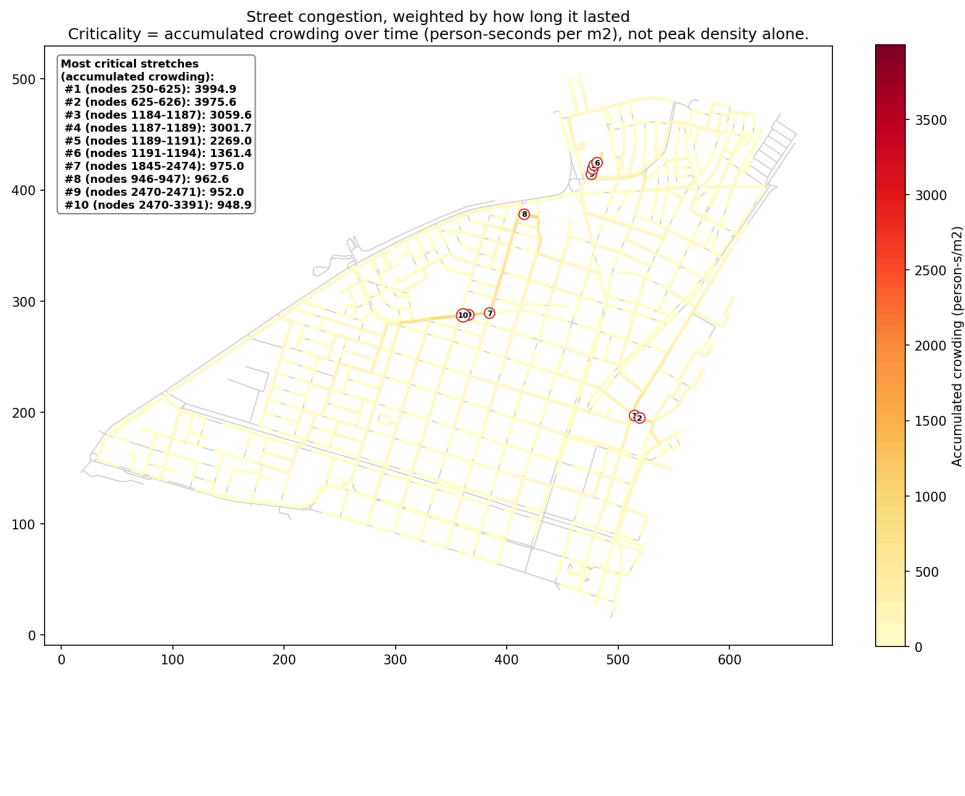


Figure 5: Streets ranked by accumulated crowding over time, with the ten most critical stretches numbered and listed.

What to look for. The ranked list names streets by the pair of intersections at their ends. These are the stretches where widening, clearing obstacles or signposting an alternative would save the most time.

Criticality here is **accumulated** crowding, not the single worst moment. A street that was briefly very busy ranks below one that was moderately crowded for many minutes — because the second one delays far more people.

2.3 The other files

File	Contents
Run_Summary.csv	One row: every headline number for this run.
Report1_Dynamics.csv	The evacuation curve, minute by minute.
Report2_Speeds.csv	Mean speed per age group over time.
Report3_Vulnerability.csv	One row per person: where they started, how long they took, and their outcome.
Report4_SafeZones.csv	Arrivals at each safe zone.
Report5_Congestion.csv	Every street, ranked by accumulated crowding.
resolved_config.txt	Every setting actually used, including defaults you did not write. Keep this: it is what makes a result reproducible.
warnings.log	Problems found in your input data. See below.

2.4 Always read `warnings.log`

Every run writes it, and the same contents appear in the PDF. It reports things GTEM found wrong with your **input data** — not with the simulation.

When there is nothing to report it says `No input warnings.` explicitly, so silence always means “checked and clean”, never “not checked”.

On the supplied Chimbote area, five warnings appear. The most serious is:

```
DISCONNECTED NETWORK: 884 of 4468 road nodes (19.8%) have no route to any safe zone.  
Anyone starting there is counted as stranded.
```

Nearly a fifth of that road network cannot reach safety at all. That is a finding about the **town**, and it deserves attention before any conclusion is drawn about behaviour or warning times.

2.5 One run is not a result

GTEM is stochastic: departure times and starting positions are drawn at random. Two runs with different seeds give different answers. Across 100 replicates of one Chimbote scenario, individual runs ranged from 80.1% to 84.5% evacuated — a spread of more than four percentage points from chance alone.

Never quote a single run

Run replicates and report the mean with its spread. Section 3 explains how, and how many replicates you need. A single run tells you what one possible evening looked like, not what the town can expect.

Part 3

Running real studies

3.1 The configuration file

Every setting lives in a plain text file. You never edit any Python file.

Copy `examples/config_example.txt`, edit the copy, and pass it with `--config`. Every line is `setting = value`; anything after `#` is a comment.

3.1.1 Settings you must provide

Setting	Example	Meaning
<code>zone</code>	<code>Chimbote_Zona1</code>	Folder name inside <code>data/</code> . Must match exactly, including capitals.
<code>adults</code>	<code>11328</code>	Number of adults to simulate.
<code>elderly</code>	<code>1917</code>	Number of elderly people.
<code>children</code>	<code>4016</code>	Number of children.
<code>tsunami_eta</code>	<code>23</code>	The single most important number. Minutes from the earthquake to the arrival of the wave. The simulation stops here.

3.1.2 Settings with sensible defaults

Setting	Default	Meaning
<code>departure_mean</code>	<code>7</code>	Average minutes before a person starts to move. The most influential behavioural assumption in the model.
<code>dt</code>	<code>10</code>	Seconds of simulated time per step. Must be above 0 and at most 10. Smaller is more accurate and slower.
<code>end_of_simulation</code>	<code>0</code>	Extra hard stop in minutes. <code>0</code> means “stop at the tsunami arrival time”, which is almost always what you want.
<code>road_width</code>	<code>2.8</code>	Usable width of one lane, in metres.
<code>capacity_multiplier</code>	<code>1</code>	Scales effective road width. <code>1</code> is undamaged; <code>0.5</code> is half.
<code>max_snap_distance</code>	<code>50</code>	How far a person may start from the road network before it is reported.
<code>vulnerability_low / _high</code>	<code>11 / 17</code>	Colour bands on the vulnerability map, in minutes.
<code>density_low / _high</code>	<code>0.3 / 3</code>	Congestion colour bands, in people per square metre.
<code>seed</code>	<code>0</code>	<code>0</code> picks a random seed and records it. Any other number reproduces that exact run.
<code>recompute_routes</code>	<code>false</code>	Rebuild the route table. Needed only after the road network changes.

Setting	Default	Meaning
<code>time_margin_analysis</code>	<code>false</code>	Measure how much more time a full evacuation would need. Roughly doubles the runtime. See Section 3.4.
<code>record_video</code>	<code>false</code>	Record an MP4. Slow, and produces a large file.

SCREENSHOT S5 — to be added

Capture: A configuration file open in a text editor (Notepad, TextEdit or VS Code), with a couple of values highlighted.

Why: Many readers have never edited a plain-text configuration file and are unsure what “edit the copy” means in practice.

Save as `figures/S5.png` and replace this box with `#figure(image("figures/S5.png"), caption: [...])` — see the appendix at the end of this manual.

3.2 When something is wrong

GTEM refuses to run rather than produce a plausible but wrong answer. Every rejection names the setting and the acceptable range:

```
Configuration error:
Invalid configuration:
- 'dt' = 30 seconds is out of range. Acceptable: 0 < dt <= 10. This is the timestep; above 10 s agents cross
several links per tick and congestion is systematically under-counted.
```

Terminal 3: A rejected configuration. Nothing is written; nothing needs cleaning up.

No invalid input ever produces an output folder that could be mistaken for a result. If a run fails part-way through, GTEM leaves a file called `FAILED.txt` in the output folder so that a half-finished folder can never be read as a finished one.

3.3 Replicates: how many runs do you need?

Because the model is stochastic, a defensible result is a **mean over replicates**. Batch runs do this for you.

Edit `examples/scenario_list.csv`. Each row is a scenario; the `Count` column says how many replicates to run:

```
Zone,Adults,Elderly,Children,TR,Count,Vuln_Low,Vuln_High,Tsunami_ETA,...
Chimbote_Zona1,11328,1917,4016,7,40,11,17,23,...
```

Then:

```
python batch_main.py --input examples/scenario_list.csv --seed 2026 --workers 4
```

`--seed` makes the whole batch reproducible. `--workers` sets how many runs happen at once; each one needs its own Java process, so start with 2–4.

3.3.1 What a batch produces

File	Contents
<code>Batch_Report.pdf</code>	Start here. Means with their spread, a comparison between scenarios, and an explicit verdict on whether you ran enough replicates.
<code>Aggregated_Summary.csv</code>	Mean, standard deviation and count per scenario.
<code>Master_Summary.csv</code>	One row per individual run.

File	Contents
Convergence.csv , Figure_Convergence.png	How the estimate settled as replicates accumulated.
Convergence_Summary.txt	The number of replicates needed for a stated precision.

3.3.2 How many is enough?

GTEM answers this from your own data rather than asking you to guess. It accumulates the coefficient of variation from run 1 onward — runs 1–2, 1–3, 1–4 and so on — and reports where it settles.

On the reference area, two independent criteria agree on about **40 replicates**:

Criterion	Replicates
Coefficient of variation stays within 10% of its final value	40
95% confidence interval of the mean within ± 0.25 percentage points	41
95% confidence interval within ± 0.50 percentage points	11
95% confidence interval within ± 0.10 percentage points	256

The batch report states plainly when a batch was **too small**, naming the scenarios whose estimates have not settled. Take that seriously: it is the difference between a result and an anecdote.

3.3.3 What an unfinished batch looks like

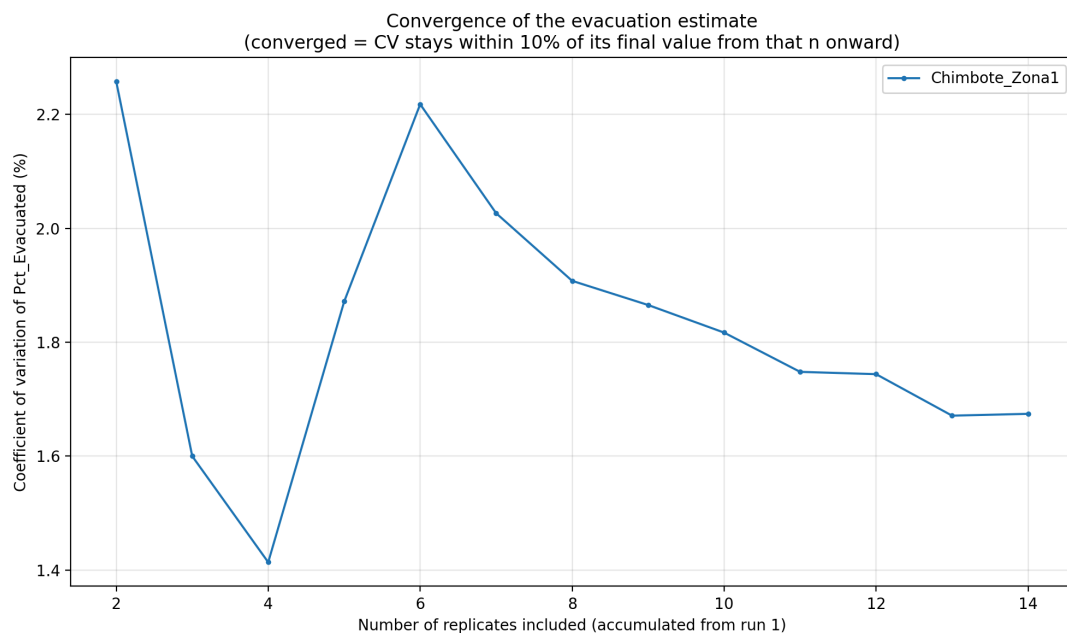


Figure 6: `Figure_Convergence.png` from a deliberately short batch of 14 replicates. The curve is still moving at the right-hand edge, and `Convergence_Summary.txt` reports CV settles at `n = not reached`. The estimate has not stabilised.

A converged curve flattens and stays flat. If yours is still rising, falling or jumping at the right-hand edge, the batch is too small — whatever the mean happens to say. In the batch above, the achieved precision was ± 0.661 percentage points, and reaching ± 0.25 would need 98 replicates.

3.4 How much time was missing?

A run bounded by the tsunami arrival time can tell you that the town ran out of time, but not by how much. Setting `time_margin_analysis = true` repeats the simulation once with the same seed and no time limit, and adds a section to the report:

On the reference area, a full evacuation needs **56.7 minutes**. The wave arrives at **23.0 minutes**. The shortfall is **33.7 minutes**.

Extra time	Reached safety	Additional people	Share
+1 min	11,846	573	3.3%
+2 min	12,295	1,022	5.9%
+3 min	12,853	1,580	9.2%
+5 min	13,862	2,589	15.0%
+10 min	15,505	4,232	24.5%

Roughly 500 people per additional minute. Because the model treats a longer warning and an earlier departure identically, this table is also a measure of what **preparedness** is worth: persuading people to leave two minutes sooner has the same effect as two minutes of extra warning.

3.5 Comparing scenarios

Comparison is the strongest use of GTEM, because both scenarios share the same assumptions and much of the uncertainty cancels out.

The supplied areas include a worked example: `Chimbote_Zona1` and `Chimbote_Zona1_colapso1`, the same town with earthquake damage to the road network. Put both in one scenario file and run them together.

Ten replicates of each, at the full population of 17,261:

Scenario	Evacuated (mean $\pm$ SD)	Stranded	Difference
Chimbote_Zona1	64.95% $\pm$ 0.72	0	baseline
Chimbote_Zona1_colapso1	29.50% $\pm$ 0.43	7,194	-35.45 points

The `Stranded` column explains the mechanism, and it is the more useful finding. Road damage does not merely slow people down: it cuts 7,194 people off from any safe zone at all. Those people cannot be helped by more warning time. They need either a route that survives the earthquake or a safe zone on their side of the damage.

How to tell a real difference from noise

Compare the difference against the standard deviations. Here the gap is 35.45 points against spreads under one point, so the difference is unmistakably real. A difference **smaller** than the spread is indistinguishable from the luck of the random seed, and should not be reported as a finding.

Note that ten replicates were enough to establish **this** difference but not to pin down either mean precisely: `Convergence_Summary.txt` for this batch asks for 32 replicates of the baseline to reach ± 0.25 percentage points. A large difference needs fewer replicates than a precise absolute number.

Part 4

Using GTEM on your own town

This is the most demanding part of the work. GTEM does not yet build a study area for you; you assemble four map layers in a GIS, and GTEM checks them.

Budget a few hours the first time. The second town is much faster.

4.1 What GTEM needs

Four shapefiles, in a folder named after your area, inside `data/` :

<code>data/My_City_Zone1/</code>		
<code>My_City_Zone1.shp</code>	zone boundary	(polygon)
<code>puntos_My_City_Zone1.shp</code>	intersections	(points)
<code>rutas_My_City_Zone1.shp</code>	road network	(lines)
<code>manzanas_My_City_Zone1.shp</code>	census blocks	(polygons)

Why some names are in Spanish

`puntos`, `rutas` and `manzanas` mean points, routes and blocks. They were kept because the project's partners already hold data under those names. They are file names only — everything inside GTEM is in English.

Each `.shp` needs its companion files (`.dbf` , `.shx` , `.prj` , `.cpg`) beside it. Copy them all.

4.2 Get the coordinate system right first

A geographic coordinate system will be rejected

Every layer must use the same **projected, metric** coordinate reference system — normally a UTM zone. GTEM measures distances in metres.

If you supply latitude and longitude (EPSG:4326, often shown as “WGS 84”), distances would be measured in **degrees** and every result would be meaningless. GTEM refuses to run rather than let that happen.

In QGIS: **Layer** ► **Export** ► **Save Features As...**, and set the CRS there.

SCREENSHOT S6 — to be added

Capture: The QGIS **Save Features As...** dialog with the CRS selector showing a UTM zone.

Why: Reprojection is the single most common preparation error, and the dialog is not obvious to a first-time user.

Save as `figures/S6.png` and replace this box with `#figure(image("figures/S6.png"), caption: [...])` — see the appendix at the end of this manual.

4.3 What each layer must contain

4.3.1 Zone boundary — `<zone>.shp`

One polygon covering the study area. GTEM uses only its extent, so a rectangle is fine. No attributes are needed.

Keep it tight: the extent sets the spatial resolution, so a boundary with a large empty margin wastes it.

4.3.2 Intersections — `puntos_<zone>.shp`

Points where roads meet, plus the safe zones. These are the nodes of the network.

Attribute	Type	Required	Meaning
<code>fid</code>	integer	yes	Unique identifier. Duplicates cause silent, seed-dependent errors.
<code>is_shelter</code>	integer	yes	<code>1</code> marks a safe zone, <code>0</code> an ordinary intersection.
<code>name</code>	text	optional	A human name such as “Colegio San Pedro”. Used to label the busiest safe zones; without it they are labelled by number.

Deciding what counts as a safe zone is your judgement, not a dataset

GTEM does not read an inundation map. It simply trusts your `is_shelter` flags. That one decision drives the entire result, so it should come from your hazard map and civil-protection plan.

At least one point must have `is_shelter = 1`, or nobody can evacuate — GTEM stops with an error rather than reporting that everyone died.

4.3.3 Road network — `rutas_<zone>.shp`

One line per street segment, joining two intersections.

Attribute	Type	Required	Meaning
<code>start_node</code>	integer	yes	<code>fid</code> of the intersection at one end.
<code>end_node</code>	integer	yes	<code>fid</code> of the intersection at the other end.
<code>cost</code>	number	yes	Length in metres , following the street rather than straight-line.
<code>lanes</code>	integer	yes	Number of lanes. Multiplied by <code>road_width</code> to give walkable width, which determines crowding. A narrow alley is <code>1</code> .

Avoid segments shorter than about one metre: a single person on a 0.1 m segment produces an absurd density and a false congestion hotspot. `check_inputs.py` counts them for you.

4.3.4 Census blocks — `manzanas_<zone>.shp`

Polygons where people start. GTEM scatters people inside them in proportion to population.

Attribute	Required	Meaning
<code>T_TOTAL</code>	strongly recommended	Resident population of the block. <code>Population</code> and <code>P0P</code> are also accepted.

Without a population attribute the results are not worth much

GTEM will still run, spreading people **uniformly** and ignoring where they actually live. It says so loudly in `warnings.log` and in the report. This is the most common preparation mistake.

Clip the blocks to the area at risk. GTEM creates people wherever you supply a block, so an unclipped layer simulates people who were never in danger.

4.4 Where to get the data

Layer	Open sources
Road network	OpenStreetMap, via the QuickOSM plugin in QGIS, Geofabrik extracts, or Overpass. Municipal cadastre if you have it.
Intersections	Derived from the road network — see below.
Census blocks	Your national statistics agency (INEI in Peru, INE in Chile, e-Stat in Japan, IBGE in Brazil, BPS in Indonesia, Eurostat in the EU). Global fallbacks: WorldPop, the Global Human Settlement Layer, or Meta's High Resolution Settlement Layer via the Humanitarian Data Exchange.
Zone boundary	Draw it yourself, or use an administrative boundary from OpenStreetMap or GADM.
Tsunami arrival time	Your national tsunami warning agency (DHN in Peru, JMA in Japan, SHOA in Chile), national hazard studies, or the NOAA historical tsunami database.
Inundation extent	National hazard maps. Failing that, a conservative elevation threshold from a free digital elevation model: Copernicus DEM GLO-30, NASADEM or ALOS World 3D.

4.5 Building the layers in QGIS

1. **Set the project coordinate system** to your metric CRS before anything else.
2. **Load a road centreline layer** — municipal data, or OpenStreetMap through the QuickOSM plugin.
3. **Clip the roads** to your study area.
4. **Split the lines at intersections**, aiming for segments of roughly 10–25 m.
5. **Extract the vertices** (**Vector** ► **Geometry Tools** ► **Extract Vertices**) and remove duplicates. This becomes `puntos_`.
6. **Add `fid`** as a unique integer on the points.
7. **Join the node identifiers to the lines**, using **Join attributes by nearest** twice — once for each end — to fill `start_node` and `end_node`.
8. **Add `cost`** with the field calculator: `$length`. Confirm the units are metres.
9. **Add `lanes`**, defaulting to 1 and raising it for main roads.
10. **Mark the safe zones**: set `is_shelter = 1` on points outside the inundation area or on vertical-evacuation buildings. Add `name` if you can.
11. **Prepare the census blocks** with a population column, clipped to the risk area.
12. **Export all four layers** into `data/<zone>/` with the required names.

SCREENSHOT S7 — to be added

Capture: QGIS **Extract Vertices** running on the road layer, with the result visible.

Why: Step 5 is where most people get lost. A picture of the menu path and the resulting point layer removes the ambiguity.

Save as `figures/S7.png` and replace this box with `#figure(image("figures/S7.png"), caption: [...])` — see the appendix at the end of this manual.

SCREENSHOT S8 — to be added

Capture: The QGIS field calculator creating `cost` with the expression `$length`.

Why: Shows both where the calculator is and what the expression looks like.

Save as `figures/S8.png` and replace this box with `#figure(image("figures/S8.png"), caption: [...])` — see the appendix at the end of this manual.

SCREENSHOT S9 — to be added

Capture: The attribute table of `puntos_` with `fid` and `is_shelter` visible, and at least one row where `is_shelter` is 1.

Why: Readers need to see that `is_shelter` is an ordinary integer column they edit by hand, not something GTEM computes.

Save as `figures/S9.png` and replace this box with `#figure(image("figures/S9.png"), caption: [...])` — see the appendix at the end of this manual.

4.6 Check before you simulate

`python check_inputs.py My_City_Zone1`

This is the step that turns a confusing mid-simulation failure into a checklist. Run it every time you change the data.

Here it is on a deliberately broken area supplied with GTEM:

```
Checking /home/you/gtem/data/Synthetic_Broken

[ OK ] zone boundary      Synthetic_Broken.shp
[ OK ] road network      rutas_Synthetic_Broken.shp
[ OK ] intersections     puntos_Synthetic_Broken.shp
[ OK ] census blocks     manzanas_Synthetic_Broken.shp

[ OK ] Synthetic_Broken.shp      CRS WGS 84 / UTM zone 17S
[ OK ] rutas_Synthetic_Broken.shp CRS WGS 84 / UTM zone 17S
[ OK ] puntos_Synthetic_Broken.shp CRS WGS 84 / UTM zone 17S
[ OK ] manzanas_Synthetic_Broken.shp CRS WGS 84 / UTM zone 17S

[ OK ] rutas_Synthetic_Broken.shp: has 'start_node'
[ OK ] rutas_Synthetic_Broken.shp: has 'end_node'
[ OK ] rutas_Synthetic_Broken.shp: has 'cost'
[ OK ] rutas_Synthetic_Broken.shp: has 'lanes'
[ OK ] puntos_Synthetic_Broken.shp: has 'fid'
[ OK ] puntos_Synthetic_Broken.shp: has 'is_shelter'
[WARN] census blocks: none of ['T_TOTAL', 'Population', 'POP'] found. Population will be spread UNIFORMLY,
ignoring where people live.

[ OK ] network: 18 nodes, 15 links, 1 safe zone(s)
[WARN] network is in 3 disconnected pieces
[FAIL] 7 of 17 starting points (41.2%) cannot reach ANY safe zone
[WARN] 1 link(s) shorter than 1 m (min 0.200 m) will produce implausible densities

Usable, with 4 warning(s). Read them before trusting results.
```

Terminal 4: `Synthetic_Broken` is included so you can see what a bad folder looks like before you meet one of your own.

4.6.1 What the messages mean

Message	What to do
geographic CRS	Reproject every layer to UTM. Nothing else matters until this is fixed.
missing <code>start_node</code>	The join in step 7 did not run, or produced a differently named column.
no safe zones	No point has <code>is_shelter = 1</code> .
X% cannot reach any safe zone	The network is fragmented, or a safe zone sits on an isolated node. Anyone starting there is counted as stranded before the simulation begins. Above roughly 10% this is worth fixing.

Message	What to do
network is in N disconnected pieces	Usually unsplit lines, or endpoints that do not quite coincide. Snap the geometry and extract the vertices again.
links shorter than 1 m	Over-splitting. Merge or delete them.
no population field	Add <code>T_TOTAL</code> to the census blocks.

4.7 A worked example to compare against

`data/Chimbote_Zona1` is a complete, working area — and a realistic one. `check_inputs.py` reports four warnings on it, including that 19.8% of its starting points cannot reach a safe zone. Compare your own folder against it rather than against an imaginary perfect dataset.

Part 5

How the model works

This part explains what happens inside a run. You do not need it to use GTEM, but you do need it to defend a result.

5.1 The sequence of a run

1. **Routes are computed once.** A single shortest-path search, seeded simultaneously from every safe zone, gives each intersection the next step towards the nearest one. The result is cached, so later runs on the same network are instant.
2. **People are placed.** Each person is put at a random point inside a census block, chosen in proportion to the population of that block, and attached to the nearest usable intersection.
3. **Each person waits.** A departure delay is drawn at random (see below).
4. **Each person walks** their route, one step per timestep, at a speed set by their age group and by how crowded the street is.
5. **The run stops** at the tsunami arrival time, or earlier if everyone has been resolved.
6. **Every person is classified** as evacuated, caught in transit, or stranded.
7. **Figures, tables and the report are written.**

5.2 Walking speed

Each age group has a free-flow speed — the speed with the street to itself:

Group	Factor	Free-flow speed
Adults	1.00	1.33 m/s
Children	0.80	1.06 m/s
Elderly	0.70	0.93 m/s

As a street becomes crowded, speed falls. Below 0.3 people per square metre everyone moves freely. Above 3.0 people per square metre movement is reduced to a crawl of 0.2 m/s. Between those two densities speed falls linearly.

The relation is scaled by each group's own free-flow speed, so the ordering between groups is preserved under congestion.

Source of the density-speed relation

Mas, E., Suppasri, A., Imamura, F. & Koshimura, S. (2015). Agent-based Simulation of the 2011 Great East Japan Earthquake/Tsunami Evacuation: An Integrated Model of Tsunami Inundation and Evacuation. *Journal of Natural Disaster Science*, 34, 41.

All six constants are configurable rather than fixed in the code.

5.3 Departure times

Nobody starts moving at the instant of the earthquake. Each person waits a random time drawn from a Rayleigh distribution whose **mean** is `departure_mean`. The distribution is right-skewed: most people leave sooner than the mean, and a tail leaves much later.

This is the assumption results are most sensitive to

The shape of the departure curve is a modelling choice, not an observation of your town. If you have survey data on how quickly people actually left during a drill or a real event, it is the single most valuable thing you can contribute to the model.

5.4 Routing

Routes are computed **before** the simulation and never change.

Every person is assumed to know the whole road network and to walk the shortest route to the **nearest safe zone by distance** — not the emptiest one, and not the fastest one under congestion.

Consequence	What it means for your results
Nobody re-routes	A person walking into a jam keeps walking into it. Congestion is somewhat overstated at the bottleneck and understated on the alternatives real people would divert to.
Perfect knowledge	Real people take familiar routes, follow crowds, go the wrong way and collect family first. GTEM's evacuation is therefore more efficient than reality . Treat clearance times as a best case.
Nearest, not emptiest	Demand concentrates. One safe zone can receive thousands while another receives nobody.

5.5 Termination and outcome accounting

The run ends at whichever comes first: everybody resolved, or the tsunami arrival time.

Every person ends in exactly one of three states, and GTEM checks while running that the three add up to the number of people simulated. If they ever did not, the run would stop with an error rather than report a number that quietly leaves people out.

5.6 What GTEM does not model

Not modelled	Consequence
Vehicles and bicycles	In a town where many people would drive, congestion is optimistic.
Safe-zone capacity	A site is counted as safe however many people arrive. Check the demand figure against real capacity yourself.
Inundation	The tsunami is one number you supply. GTEM does not simulate the wave, its extent or its depth.
Disability and reduced mobility	Only three age groups exist. A town with a large care population is not well represented.
Households and group behaviour	Everyone evacuates alone. Nobody waits for or collects anybody.
Time of day	The population is a night-time, everyone-at-home estimate. Schools, workplaces and beaches are not represented.
Buildings, debris and injury	Nothing obstructs movement except other people.

5.7 The state of validation

GTEM has not been validated, and the one benchmark available says it is optimistic

Everything above describes what GTEM **computes**. Whether that matches a real evacuation is a separate and harder question.

GTEM descends from TUNAMI-EVAC1 (Mas, 2012), which **was** validated against the 2011 evacuation of Arahama, Sendai: about **90% of 2,271 people saved**, and 520 sheltered in the evacuation building. Under matched assumptions the ancestor gives **81.5%** and GTEM **83.4%** — close to each other, both below the observed figure.

That is a starting point, not a validation: one scenario, one seed, one aggregate number. The comparison is also acutely sensitive to the departure assumption — the same GTEM run with a 26-minute-earlier mean departure evacuates **100%**. [validation/README.md](#) sets this out in full.

Until a real validation exists: use GTEM to compare options, state the assumptions, and say in any report that the model is unvalidated.

Part 6

Reference

6.1 Writing the report in Spanish

The figures and both PDF reports can be produced in Spanish. Either set it in the configuration file:

```
language = es
```

or override it on the command line, which is convenient when you want the same scenario in both languages:

```
python main.py --config my_run.txt --language es
```

`batch_main.py` takes the same option.

Translated	Always English
Figure titles, axis labels and legends; every heading, table and sentence in the single-run and batch PDF reports.	Configuration keys, CSV column headers, output file names, and the settings listed in the report's configuration table.

Why the data tables stay in English

So that two runs can be compared column by column whatever language their reports were written in, and so a script written against your results keeps working when a colleague runs the model in the other language. The Spanish report says this on the page where it matters.

Two parts of a Spanish report remain in English: the warnings raised by the simulation engine, and the console output. The report states this where the warnings appear, so a reader is not left wondering.

To add another language, copy the `EN` table in `src/text_strings.py`, translate the values, and register it in `TABLES`. The test suite checks that every language has the same keys and the same `{placeholders}` as English.

6.2 Command reference

```
usage: main.py [-h] --config CONFIG [--output OUTPUT] [--language {en,es}]

GTEM 1.0.0 - run one evacuation simulation.

options:
  -h, --help            show this help message and exit
  --config CONFIG        Path to the run configuration file (see
                        examples/config_example.txt).
  --output OUTPUT        Output folder. Default: Outputs/<zone>/<seed>/
  --language {en,es}    Language of the figures and the PDF report. Overrides the
                        'language' setting in the configuration file. Default: en.
```

Terminal 5: `python main.py --help`

```
usage: batch_main.py [-h] [--input INPUT] [--output OUTPUT] [--workers WORKERS]
                  [--memory-mb MEMORY_MB] [--seed SEED] [--reports]
                  [--language {en,es}]

GTEM 1.0.0 - run GTEM scenarios in batch.

options:
  -h, --help            show this help message and exit
  --input INPUT          CSV table of scenarios.
  --output OUTPUT        Folder name inside Outputs/ (default: timestamp).
  --workers WORKERS      Concurrent JVMs. Each uses up to --memory-mb.
  --memory-mb MEMORY_MB Maximum RAM per JVM.
  --seed SEED            Master seed. Makes the whole batch reproducible.
  --reports              Also write figures and a PDF for every run (slower).
  --language {en,es}    Language of the figures and the PDF reports. Default: en.
```

Terminal 6: `python batch_main.py --help`

Command	Purpose
<code>python check_environment.py</code>	Is this machine able to run GTEM?
<code>python check_inputs.py <zone></code>	Is this area's data usable?
<code>python main.py --config <file></code>	Run one simulation.
<code>python main.py --config <file> --language es</code>	The same run, reported in Spanish.
<code>python batch_main.py --input <csv></code>	Run many, with statistics.
<code>python -m pytest tests/ -m "not engine"</code>	Fast self-test, about a minute.
<code>python -m pytest tests/</code>	Full self-test, about half an hour.

6.3 Exit codes

Code	Meaning
0	Finished; all outputs written.
1	The run failed. A <code>FAILED.txt</code> marker is left in the output folder.
2	The configuration or the input data is invalid. Nothing was run.

6.4 Folder layout

```

CHANGELOG.md
CITATION.cff
LICENSE
Outputs
README.md
__pycache__
assets
batch_main.py
benchmarks
cache
check_environment.py
check_inputs.py
data
docs
environment.yml
examples
main.py
requirements-tools.txt
requirements.txt
src
tests
tools
validation

```

Terminal 7: The top level of a GTEM folder.

6.5 Troubleshooting

Symptom	Cause and cure
JVM DLL not found — but the file is there	Python and NetLogo's Java runtime are built for different processors. On an Apple Silicon Mac this usually means an Intel build of Python, often Anaconda under <code>/opt/anaconda3</code> , running under Rosetta. Check with <code>python -c "import platform; print(platform.machine())"</code> it must print <code>arm64</code> . Recreate the environment with Miniforge if it does not. <code>check_environment.py</code> now reports this before you run anything.
No NetLogo installation was found	NetLogo is missing or installed elsewhere. Install 7.0.4, or set <code>NETLOGO_HOME</code> to its folder. The error lists the places GTEM looked.
A package is reported missing	You have not activated the environment. Run <code>conda activate gtem</code> in this terminal.
Zone folder not found	The <code>zone</code> setting does not match a folder inside <code>data/</code> . Capitals matter.
uses a geographic (lat/lon) CRS	Reproject every layer to a metric CRS such as UTM.
is out of range	The message names the setting and its acceptable values. Nothing was run.
The run seems to hang on macOS or Linux	Should not occur in this version, which starts the Java engine in headless mode. If it does, report it — include your operating system version.

Symptom	Cause and cure
Results differ between two identical runs	Check that <code>seed</code> is not <code>0</code> . A seed of <code>0</code> deliberately picks a new random seed each time, and records it in the results.
Errors mentioning very long file paths on Windows	Move the GTEM folder somewhere with a short path, such as <code>C:\GTEM</code> .

6.6 Glossary

Term	Meaning
Agent	One simulated person.
Caught in transit	Still walking when the wave arrived.
Coefficient of variation	The spread of a set of results divided by their mean, as a percentage. Used to judge whether enough replicates were run.
CRS	Coordinate reference system. GTEM requires a projected, metric one.
ETA	Estimated time of arrival — of the tsunami, in minutes.
Free-flow speed	Walking speed with the street to yourself.
Replicate	One run of the same scenario with a different random seed.
Safe zone	A point flagged <code>is_shelter = 1</code> , where people are counted safe.
Seed	The number that fixes the random draws, making a run repeatable.
Stranded	No route to any safe zone from the starting point.
Timestep (<code>dt</code>)	Seconds of simulated time advanced in each step.

6.7 Licence, citation and credits

GTEM is released under the MIT licence. The study-area data supplied with it is released for free redistribution alongside the software.

Authors. Erick Mas (IRIDeS, Tohoku University, Japan), Luis Moya (Pontificia Universidad Catolica del Peru), Jheyder Perez (Pontificia Universidad Catolica de Chile).

GTEM derives from TUNAMI-EVAC (github.com/erick2307/TUNAMI-EVAC). Funded by the Coalition for Disaster Resilient Infrastructure under the CDRI Fellowship Programme 2025–2026. Built on NetLogo (Wilensky, 1999).

If you use GTEM, please cite both the software — see `CITATION.cff` — and the source of the density–speed relation, Mas et al. (2015).

6.8 Further reading in the GTEM folder

Document	Contents
<code>docs/LIMITATIONS.md</code>	The full statement of what GTEM can and cannot answer. Read before presenting results.
<code>docs/PREPARING_YOUR_CITY.md</code>	The input schema in reference form.
<code>docs/VERIFICATION.md</code>	Evidence that the model computes what it claims.

Document	Contents
<code>docs/ROADMAP.md</code>	Known gaps and what is planned.
<code>validation/README.md</code>	Why GTEM is not yet validated, and what that needs.
<code>CHANGELOG.md</code>	Release history.

Appendix — screenshots still to be captured

This manual is written so that it is complete and usable **without** screenshots. Nine places would nonetheless be clearer with one, and each is marked in the text by an orange dashed box.

To add one: take the screenshot, save it in `docs/manual/figures/` under the filename below, and in `manual.typ` replace the whole `#todo-shot(...)` block with

```
#figure(image("figures/S1.png", width: 90%),
        caption: [NetLogo 7.0.4 on the downloads page.]
```

Then recompile:

```
typst compile docs/manual/manual.typ docs/manual/GTEM_Manual.pdf
```

No.	Filename	What to capture
S1	S1.png	The NetLogo downloads page with version 7.0.4 visible. Readers often install the newest version instead.
S2	S2.png	The unzipped GTEM folder in Windows Explorer or macOS Finder, showing <code>main.py</code> , <code>data</code> , <code>examples</code> , <code>docs</code> and <code>src</code> at the top level.
S3	S3.png	A Miniforge Prompt immediately after <code>conda activate gtem</code> , with the <code>(gtem)</code> prefix at the start of the line.
S4	S4.png	The <code>Outputs/Synthetic_Corridor/1/</code> folder showing all 14 output files.
S5	S5.png	A configuration file open in a plain-text editor (Notepad, TextEdit or VS Code), so readers see the <code>key = value</code> layout.
S6	S6.png	The QGIS Save Features As... dialog with the CRS selector showing a UTM zone. Reprojection is the commonest preparation error.
S7	S7.png	QGIS Vector ► Geometry Tools ► Extract Vertices , used to derive the intersection points from the road network.
S8	S8.png	The QGIS field calculator creating the <code>cost</code> field from <code>\$length</code> .
S9	S9.png	The intersections attribute table with the <code>is_shelter</code> column, showing both <code>0</code> and <code>1</code> values.

Take every screenshot at a comfortable reading size, on a light background if possible, and crop to the dialog or window concerned rather than the whole desktop.